

IOCL GRADE-A 2026

CS/IT WORKBOOK

DAY 1 — DBMS FOUNDATIONS

19 AUGUST 2026

Topic: Relational Model, Schema/Instance, Keys, Constraints, ER Basics
+ Exam Extension: Functional Dependencies, Closure & Keys

Target session length: $\approx$ 2 hours 15 minutes (core) + 45–60 minutes (exam extension, optional)
Exam date: 24 September 2026
Study date: 19 August 2026

Goal: Comfortable CBT clearance – high-yield E/M level MCQs, not GATE-depth mastery. The exam extension adds only the specific FD/closure/key patterns that recur in PSU and GATE papers – not a full normalization chapter.

Session Plan

- 55–65 min – Concepts, mental models, worked examples
- 40–50 min – Guided + independent MCQ practice
- 15–20 min – Timed mini-test (10Q)
- 10–15 min – Error log + revision checklist + cheat sheet
- **45–60 min (optional, exam-focused)** – Day 1 Exam Extension: closure, candidate keys, superkey counting

This is a study diagnostic tool, not an official IOCL cutoff predictor.

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1 Learning Objectives

By the end of this session you should be able to:

- Distinguish DBMS, RDBMS, database, schema, and instance.
- Identify super key, candidate key, primary key, foreign key, and composite key from a given table.
- Distinguish entity integrity, referential integrity, and domain constraints.
- Reason correctly about NULL behavior (arithmetic, comparisons, foreign keys).
- Read a short ER description and identify entities, relationships, and cardinality (1:1, 1:N, M:N).
- Answer common direct-fact and small-scenario DBMS MCQs quickly and confidently.
- **(Exam Extension)** Calculate attribute closure, find candidate keys mechanically, and count superkeys correctly whether a relation has one, two, or more candidate keys.

Why this matters for IOCL CBT

DBMS foundational theory (not SQL syntax) is a classic source of quick, high-confidence marks in CS/IT technical sections of PSU exams like IOCL. These are typically direct-recall or one-step reasoning questions – exactly the kind that reward a focused 2-hour session rather than deep GATE preparation. Your existing SQL/PostgreSQL workbook already covers query syntax; today builds the conceptual layer underneath it. The exam extension exists because closure/candidate-key/superkey-counting questions have recently appeared verbatim in GATE CSE/DA papers and are increasingly common in PSU-pattern papers too – they are cheap marks once the mechanical steps are memorized.

2 Concepts and Mental Models

2.1 1. DBMS vs RDBMS

What: A DBMS (Database Management System) is software that lets you create, store, retrieve, update, and manage data. An RDBMS (Relational DBMS) is a DBMS that organizes data strictly as relations (tables) with rows and columns, and enforces relational integrity rules.

Why it exists: Before DBMS, applications managed flat files directly – leading to redundancy, inconsistency, poor concurrent access, and no standard query mechanism. A DBMS centralizes control.

How it works: The RDBMS additionally guarantees each table has a defined schema, enforces keys/constraints, and typically supports SQL as the query language.

Feature	DBMS	RDBMS
Data structure	Files, tables, hierarchical, or navigational	Strictly tables (relations)
Relationships between data	Not necessarily enforced	Enforced via keys (PK/FK)
Normalization	Not mandatory	Supported / expected
Multiple users / ACID	Varies	Typically yes (ACID transactions)
Examples	File systems, some NoSQL stores, DBMS in general	MySQL, PostgreSQL, Oracle, SQL Server

Common Trap

Exam trap: “Every RDBMS is a DBMS, but not every DBMS is an RDBMS.” Many MCQs test this one-directional relationship. Also: a plain file system is NOT a DBMS (no query language, no integrity enforcement, no concurrency control).

2.2 2. Database, Schema, and Instance

Database: The actual collection of stored, related data at any point in time.

Schema: The overall logical design / structure of the database – table names, column names, data types, constraints. The schema is defined once and changes rarely.

Instance: The data actually present in the database at a given moment in time. The instance changes frequently (every INSERT/UPDATE/DELETE creates a new instance).

Mental Model

Think of schema as the blueprint of a house (fixed), and instance as who is inside the house right now (changes constantly). The blueprint (schema) doesn't change every time someone walks in or out (instance).

Common Trap

Exam trap: Questions often swap “schema” and “instance” definitions as distractors. Remember: Schema = structure/design (intension); Instance = data snapshot (extension).

2.3 3. Relation, Tuple, Attribute, Domain

Term	Formal meaning	Everyday equivalent
Relation	A table (a set of tuples over the same attributes)	The table itself
Tuple	A single ordered set of attribute values	A row
Attribute	A named column of the relation	A column
Domain	The set of allowable atomic values an attribute can take	Valid “type/range” of a column (e.g., Age: positive integers)

Example table – Student:

RollNo	Name	Age	Dept
101	Asha	20	CS
102	Bala	21	IT
103	Chetan	20	CS

Here the relation is *Student*; each row is a tuple; RollNo, Name, Age, Dept are attributes; the domain of Age could be defined as integers between 17 and 30.

Common Trap

Exam trap: “Degree” of a relation = number of attributes (columns); “Cardinality” of a relation = number of tuples (rows). These two terms are frequently swapped in distractor options.

2.4 4. The Key Hierarchy

Why keys exist: Keys exist to uniquely identify each tuple in a relation, enabling unambiguous referencing, enforcing entity integrity, and supporting relationships between tables (via foreign keys).

Key type	Definition
Super key	Any set of attributes that uniquely identifies a tuple (may contain extra/redundant attributes).
Candidate key	A minimal super key – no attribute can be removed without losing uniqueness.
Primary key	One candidate key chosen by the designer to be the main identifier. Cannot be NULL.
Alternate key	Candidate key(s) not chosen as the primary key.
Composite key	A key (super/candidate/primary) consisting of two or more attributes together.
Foreign key	An attribute (or set) in one relation that refers to the primary key of another (or the same) relation, enforcing referential integrity.
Natural key	A key made of real-world/business data already present (e.g., Aadhaar number).
Surrogate key	An artificial key with no business meaning, generated by the system (e.g., auto-increment ID).

Mental Model

Every Primary Key is a Candidate Key. Every Candidate Key is a Super Key. Not the reverse.

$$\text{Super Keys} \supseteq \text{Candidate Keys} \supseteq \{\text{Primary Key}\}$$

Worked example – “given this table, identify the key”:

RollNo	Aadhaar	Name	Email
101	9111-2222-3333	Asha	asha@x.com
102	9444-5555-6666	Bala	bala@x.com
103	9777-8888-9999	Chetan	chetan@x.com

- Candidate keys: {RollNo}, {Aadhaar}, {Email} – each alone is unique.
- Super keys: all of the above, plus any combination such as {RollNo, Name}, {RollNo, Aadhaar, Email}, etc.
- Primary key (chosen): say {RollNo}.
- Alternate keys: {Aadhaar}, {Email}.
- Composite key example: if no single column were unique, e.g. {Dept, SeatNo} together could form a composite candidate key.

Common Trap

Exam trap 1: “Every super key is a candidate key” – FALSE. It’s the reverse.

Exam trap 2: A composite key means no single attribute in that combination is enough alone; each part may repeat individually, only the combination is unique.

Exam trap 3: A relation can have multiple candidate keys but exactly one primary key.

2.5 5. Integrity Constraints

Constraint	Rule
Domain constraint	Every attribute value must come from its defined domain (correct data type/range).
Entity integrity	Primary key attribute(s) can never be NULL, and must be unique per tuple.
Referential integrity	A foreign key value must either match an existing primary key value in the referenced relation, or be NULL (if allowed).
Key / uniqueness constraint	No two tuples can have the same value for a candidate key.

Mental Model

Entity integrity protects the identity of a row (PK never NULL). Referential integrity protects the links between tables (FK must point to something real, or be NULL).

Common Trap

Exam trap: A foreign key value CAN be NULL (if the relationship is optional) – referential integrity is violated only when the FK holds a non-NULL value that does not exist in the referenced table’s primary key column. Also: deleting a referenced primary-key row while dependent foreign keys exist violates referential integrity unless cascading rules are applied.

2.6 6. NULL vs Zero vs Empty String

Value	Meaning
NULL	Unknown / not applicable / missing – it is not a value, not even “blank”.
0 (zero)	A known, definite numeric value.
” (empty string)	A known, definite value – a string of length zero.

Common Trap

Exam trap: NULL = NULL evaluates to NULL (unknown), not TRUE. Any arithmetic or comparison involving NULL yields NULL (three-valued logic: TRUE / FALSE / UNKNOWN). Also, COUNT(column) skips NULLs but COUNT(*) counts all rows regardless of NULLs. NULL $\neq$ 0 and NULL $\neq$ ”.

2.7 7. ER Model Basics

What/Why: The Entity-Relationship (ER) model is a conceptual design tool used before building the relational schema, to visually capture real-world entities, their attributes, and how they relate.

Term	Meaning
Entity	A real-world object/concept with independent existence (e.g., Student, Course).
Attribute	A property describing an entity (e.g., Student has Name, RollNo).
Relationship	An association among two or more entities (e.g., Student Enrolls in Course).
Cardinality	The number of instances of one entity that can/must associate with instances of another.

Cardinality types:

- **1:1 (one-to-one):** Each entity instance associates with at most one instance of the other (e.g., Employee–Parking Slot).
- **1:N (one-to-many):** One instance of entity A associates with many instances of entity B, but each B relates to only one A (e.g., Department–Employee).
- **M:N (many-to-many):** Instances on both sides can relate to multiple instances on the other side (e.g., Student–Course).

Common Trap

Exam trap: When converting ER to relational schema, a M:N relationship always requires a separate junction/bridge table holding foreign keys from both entities; a 1:N relationship is normally captured by placing the foreign key on the “N” side (no extra table needed).

2.8 8. Bridge to Functional Dependencies (preview only)

Just the intuition – depth comes in the Exam Extension (Section 10) below.

A functional dependency $X \rightarrow Y$ means: if two tuples agree on all attributes in X , they must also agree on all attributes in Y . In other words, the value(s) of X determine the value(s) of Y .

Example: In a Student table, RollNo $\rightarrow$ Name (each roll number determines exactly one name).

This is the seed idea for normalization. Today, in the core session, you only need to recognize the notation $X \rightarrow Y$ and read it as “ X determines Y ”. The optional Exam Extension (Section 10) builds directly on this with closure, candidate-key finding, and superkey counting – the exact patterns tested in recent GATE/PSU papers.

3 Worked MCQs (with full explanations)

Study these 8 questions carefully before attempting the guided/independent sets. Each is representative of a real recurring PSU/GATE-style pattern.

W1 [DBMS] [Key Hierarchy] [E] [Definition]

Which of the following statements is TRUE?

- (a) Every candidate key is a primary key. (b) Every super key is a candidate key.
(c) Every primary key is a candidate key. (d) A relation can have more than one primary key.

Answer

(c). A primary key is simply the candidate key chosen by the designer, so it is always a candidate key. (a) is false – a table can have multiple candidate keys but only one is chosen as PK. (b) is false – super keys can contain redundant attributes; only minimal super keys are candidate keys. (d) is false – exactly one PK per relation.

Trap: Confusing the direction of the “every X is a Y” relationship among super/candidate/primary keys.

W2 [DBMS] [Schema/Instance] [E] [Definition]

The overall design of a database that rarely changes is called its:

- (a) Instance (b) Schema (c) Snapshot (d) Extension

Answer

(b) Schema. Schema = the fixed logical structure (also called intension). (a) and (d) refer to the current data (extension), which changes often. (c) is not a standard DBMS term used this way.

W3 [DBMS] [Integrity] [M] [Application]

A foreign key column in table Orders references the primary key of table Customers. Which action is guaranteed to violate referential integrity (assuming no cascade rules)?

- (a) Inserting an Orders row with FK = NULL.
(b) Deleting a Customers row that has zero matching Orders rows.
(c) Inserting an Orders row whose FK value does not exist in Customers.
(d) Updating a Customers primary key value that no Orders row references.

Answer

(c). Referential integrity requires every non-NULL FK value to exist in the referenced PK column. (a) is allowed (NULL FK = “no relationship yet”, permitted unless FK is also marked NOT NULL). (b) and (d) are safe because no dependent rows are affected.

W4 [DBMS] [ER Model] [M] [Scenario]

In an ER diagram, “One Department has many Employees, but each Employee belongs to exactly one Department” describes which cardinality, and where would the foreign key be placed?

- (a) M:N; junction table needed. (b) 1:1; FK on either side.
(c) 1:N; FK placed on the Employee (the “many”) side. (d) 1:N; FK placed on the Department (the “one”) side.

Answer

(c). This is a 1:N relationship. The foreign key always goes on the “many” side (Employee), storing the Department’s primary key.

W5 [DBMS] [NULL] [M] [Conceptual]

If `col1` contains NULL, what is the result of the expression `col1 = col1`?

- (a) TRUE (b) FALSE (c) NULL / UNKNOWN (d) Error, cannot be evaluated

Answer

(c). NULL represents an unknown value, so comparing an unknown to itself is still unknown (three-valued logic).

W6 [DBMS] [Relational Terms] [E] [Definition]

The number of tuples in a relation is called its:

- (a) Degree (b) Cardinality (c) Domain (d) Arity

Answer

(b) Cardinality. Degree/Arity = number of attributes (columns), not rows.

W7 [DBMS] [Keys] [M] [Small Table]

Table Booking(FlightNo, Date, SeatNo, PassengerID) – a passenger can only book one seat on a given flight on a given date, and each seat on a flight+date is occupied by only one passenger. Which is the most likely composite candidate key?

- (a) {FlightNo} (b) {FlightNo, Date, SeatNo} (c) {PassengerID} (d) {Date}

Answer

(b). No single column is unique alone, but the combination FlightNo+Date+SeatNo uniquely identifies one booking.

W8 [DBMS] [DBMS vs RDBMS] [E] [Standard Fact]

Which of these is NOT a required property of a strict relational database, per Codd's model, but IS commonly expected in modern general-purpose DBMS software?

- (a) Enforcement of primary key uniqueness. (b) Support for tabular (row/column) storage.
(c) Guaranteed sub-millisecond query response time. (d) Support for foreign key constraints.

Answer

(c). Performance guarantees like response time are implementation/hardware-dependent, not a defining relational property.

4 Guided MCQs (10 Questions)

Attempt these with the concepts fresh. Answers appear only in the Answer Key section at the end.

1. [DBMS] [DBMS/RDBMS] [E] [Definition] A plain spreadsheet file (e.g., a CSV) is best described as:
(A) A DBMS (B) An RDBMS (C) Neither a DBMS nor RDBMS (D) Both a DBMS and RDBMS
2. [DBMS] [Schema/Instance] [E] [Definition] The current set of rows actually stored in a table right now is called the:
(A) Schema (B) Instance (C) Domain (D) Relation scheme
3. [DBMS] [Keys] [E] [Definition] A key that consists of two or more attributes together is called a:
(A) Candidate key (B) Composite key (C) Surrogate key (D) Alternate key
4. [DBMS] [Keys] [M] [Comparison] Which statement correctly relates super keys and candidate keys?
(A) Candidate keys are a subset of super keys (B) Super keys are a subset of candidate keys
(C) They are always equal in number (D) They are unrelated concepts
5. [DBMS] [Integrity] [E] [Definition] The rule that a primary key value can never be NULL is called:
(A) Referential integrity (B) Domain constraint (C) Entity integrity (D) Key constraint only
6. [DBMS] [Integrity] [M] [Scenario] A foreign key in table B references the primary key of table A. Table A's row is deleted, but no cascade rule exists and dependent rows exist in B. What happens?
(A) Deletion succeeds silently (B) Deletion is rejected due to referential integrity (C) B's rows are auto-deleted (D) B's FK becomes 0
7. [DBMS] [NULL] [M] [Conceptual] $5 + \text{NULL}$ evaluates to:
(A) 5 (B) 0 (C) NULL (D) Error
8. [DBMS] [ER Model] [E] [Definition] In ER modeling, a real-world object with independent existence (like "Employee") is called a(n):
(A) Attribute (B) Relationship (C) Entity (D) Cardinality
9. [DBMS] [ER Model] [M] [Scenario] "A Student can enroll in many Courses, and a Course can have many Students" is an example of:
(A) 1:1 (B) 1:N (C) M:N (D) None of these
10. [DBMS] [FD Bridge] [E] [Notation] $X \rightarrow Y$ is read as:
(A) Y determines X (B) X and Y are independent (C) X determines Y (D) X equals Y

5 Independent MCQs (15 Questions)

Attempt without referring back to the concept section. Tag legend: [E]=Easy [M]=Medium [H]=Hard.

1. [DBMS] [DBMS/RDBMS] [E] [Standard Fact] Which of these is a defining feature of an RDBMS but not guaranteed in every general DBMS?
(A) Data storage on disk (B) Enforcement of relational keys and constraints (C) Ability to store text (D) Use of a computer
2. [DBMS] [Schema/Instance] [M] [Comparison] Which pair correctly matches “changes rarely” vs “changes frequently”?
(A) Instance / Schema (B) Schema / Instance (C) Domain / Attribute (D) Tuple / Relation
3. [DBMS] [Relational Terms] [E] [Definition] The number of attributes in a relation is its:
(A) Cardinality (B) Degree (C) Domain (D) Instance
4. [DBMS] [Relational Terms] [M] [Small Table] Table Emp(EmpID, PAN, Name, Salary) where EmpID and PAN are both individually unique. How many candidate keys exist (ignoring combinations)?
(A) 1 (B) 2 (C) 3 (D) 4
5. [DBMS] [Keys] [M] [Small Table] In the table above, if PK = EmpID, then PAN is called a(n):
(A) Foreign key (B) Alternate key (C) Composite key (D) Surrogate key
6. [DBMS] [Keys] [H] [Trap] A surrogate key is best described as:
(A) A key with real-world business meaning (B) A system-generated key with no business meaning (C) A key that must be composite (D) A key that can be duplicated
7. [DBMS] [Integrity] [E] [Definition] Ensuring every attribute value belongs to its predefined valid set of values enforces:
(A) Entity integrity (B) Referential integrity (C) Domain constraint (D) Key constraint
8. [DBMS] [Integrity] [M] [Scenario] Which of these values is permitted in a foreign key column, assuming the FK is nullable?
(A) Any value not present in the referenced table (B) NULL (C) A negative ID that was never assigned (D) A duplicate of another row’s non-matching value
9. [DBMS] [NULL] [E] [Conceptual] Which of these correctly represents “unknown” rather than a real value?
(A) 0 (B) ” (empty string) (C) NULL (D) -1
10. [DBMS] [NULL] [M] [Conceptual] COUNT(*) vs COUNT(column with nulls) on a 10-row table where 3 rows have NULL in that column returns:
(A) 10 and 10 (B) 10 and 7 (C) 7 and 10 (D) 7 and 7
11. [DBMS] [ER Model] [E] [Definition] A property describing an entity (such as Name or Age) is called an:
(A) Entity (B) Attribute (C) Relationship (D) Tuple
12. [DBMS] [ER Model] [M] [Scenario] “Each Passport belongs to exactly one Citizen, and each Citizen has at most one Passport” is:
(A) 1:1 (B) 1:N (C) M:N (D) Undefined

13. [DBMS] [ER Model] [M] [Conversion] An M:N relationship between two entities is converted into a relational schema by:
(A) Placing FK on either entity table (B) Merging both entities into one table (C) Creating a separate junction table with FKs to both entities (D) Ignoring the relationship
14. [DBMS] [FD Bridge] [M] [Notation] If RollNo $\rightarrow$ Name holds, which statement is correct?
(A) Two tuples with the same Name must have the same RollNo (B) Two tuples with the same RollNo must have the same Name (C) RollNo and Name are unrelated (D) Name determines RollNo
15. [DBMS] [Mixed] [H] [Trap] Which of the following is FALSE?
(A) A relation can have only one primary key (B) A relation can have multiple candidate keys
(C) A composite key must always include the primary key (D) A foreign key can reference a primary key in the same table (self-reference)

6 Timed Mini-Test (10 Questions – 15 to 20 minutes)

Instructions: Set a timer for 15–20 minutes. Attempt all 10 without pausing. Do not check the answer key until finished.

1. [DBMS] [DBMS/RDBMS] [E] [Definition] An RDBMS differs from a generic DBMS mainly because it:
(A) Runs faster (B) Organizes data strictly as related tables with enforced constraints (C) Cannot store images (D) Requires no schema
2. [DBMS] [Schema/Instance] [E] [Definition] Which changes every time a row is inserted?
(A) Schema (B) Instance (C) Domain (D) Attribute name
3. [DBMS] [Relational Terms] [E] [Definition] A single row of a relation is formally called a:
(A) Attribute (B) Tuple (C) Domain (D) Schema
4. [DBMS] [Keys] [M] [Small Table] Table Vehicle(RegNo, ChassisNo, OwnerName), both RegNo and ChassisNo are unique. If RegNo is chosen as PK, ChassisNo is a(n):
(A) Foreign key (B) Composite key (C) Alternate key (D) Surrogate key
5. [DBMS] [Integrity] [M] [Scenario] A NOT NULL, UNIQUE constraint on a column that is NOT the primary key enforces behavior similar to:
(A) A candidate key (B) A foreign key (C) Referential integrity (D) Domain constraint only
6. [DBMS] [NULL] [E] [Conceptual] NULL is best described as:
(A) Equivalent to zero (B) Equivalent to empty string (C) Absence/unknown of a value (D) A syntax error
7. [DBMS] [ER Model] [E] [Definition] The association between two or more entities is called a:
(A) Attribute (B) Relationship (C) Domain (D) Instance
8. [DBMS] [ER Model] [M] [Scenario] “One Author can write many Books, but each Book has exactly one Author” – where should the foreign key go?
(A) Author table (B) Book table (C) A separate junction table (D) Either table
9. [DBMS] [FD Bridge] [E] [Notation] In $X \rightarrow Y$, X is called the:
(A) Dependent (B) Determinant (C) Candidate (D) Domain
10. [DBMS] [Mixed] [M] [Trap] Which statement is TRUE?
(A) A table can have zero candidate keys (B) A table can have multiple primary keys (C) A foreign key value must always be non-NULL (D) A primary key value must always be non-NULL

7 End-of-Day Quiz (5 Questions)

Quick final check – these mix easy recall with one small scenario.

1. [DBMS] [DBMS/RDBMS] [E] [Definition] File systems lack which core DBMS capability?
(A) Storing data on disk (B) Enforced integrity constraints and query language (C) Allowing folders (D) Backup capability
2. [DBMS] [Keys] [E] [Definition] The minimal super key is called a:
(A) Composite key (B) Candidate key (C) Surrogate key (D) Alternate key
3. [DBMS] [Integrity] [M] [Scenario] Which constraint is violated if an Orders.CustomerID references a CustomerID that does not exist in the Customers table?
(A) Domain constraint (B) Entity integrity (C) Referential integrity (D) None, this is allowed
4. [DBMS] [ER Model] [E] [Definition] The number of entity instances that can participate in a relationship defines its:
(A) Attribute (B) Cardinality (C) Domain (D) Schema
5. [DBMS] [FD Bridge] [M] [Notation] If every RollNo maps to exactly one Name in a table, we write:
(A) $\text{Name} \rightarrow \text{RollNo}$ (B) $\text{RollNo} \rightarrow \text{Name}$ (C) $\text{RollNo} = \text{Name}$ (D) $\text{Name} \leftrightarrow \text{RollNo}$

8 Complete Answer Key

8.1 Guided MCQs – Answers

1. (C) Neither. A CSV has no enforced schema, constraints, or query engine – it's just a file.
2. (B) Instance. The current stored rows = instance; schema is the fixed design.
3. (B) Composite key. By definition, spans two or more attributes.
4. (A) Candidate keys are a subset of super keys.
5. (C) Entity integrity. Governs primary key non-nullability and uniqueness.
6. (B) Deletion is rejected. Without a cascade rule, referential integrity blocks deleting a referenced row that still has dependents.
7. (C) NULL. Any arithmetic operation involving NULL propagates NULL.
8. (C) Entity. Attributes describe entities; relationships connect entities.
9. (C) M:N. Both sides can have multiple associated instances.
10. (C) X determines Y. Core reading of a functional dependency.

8.2 Independent MCQs – Answers

1. (B). Enforced relational keys/constraints define RDBMS specifically.
2. (B) Schema / Instance. Schema changes rarely; instance changes frequently.
3. (B) Degree. Degree = number of columns.
4. (B) 2. Both EmpID and PAN individually qualify as separate candidate keys.
5. (B) Alternate key. Any candidate key not chosen as PK.
6. (B). Surrogate keys are artificial, system-generated identifiers.
7. (C) Domain constraint. Values fall within the attribute's valid domain/type/range.
8. (B) NULL. A nullable FK may legitimately hold NULL.
9. (C) NULL. 0 and " are both definite, known values.
10. (B) 10 and 7. COUNT(*) counts all rows; COUNT(column) skips NULL rows.
11. (B) Attribute. Attributes are properties of entities.
12. (A) 1:1. Both directions are capped at "at most one".
13. (C) Junction/bridge table. M:N relationships need a separate linking table.
14. (B). $X \rightarrow Y$ means tuples agreeing on X must agree on Y.
15. (C) is FALSE (the answer to select). A composite key does NOT have to include the primary key.

8.3 Timed Mini-Test – Answers

1. (B). Enforced relational structure and constraints are the defining RDBMS trait.
2. (B) Instance. Every INSERT changes the current data snapshot.
3. (B) Tuple. A row = tuple.
4. (C) Alternate key. ChassisNo is a candidate key not chosen as PK.
5. (A) A candidate key. A NOT NULL + UNIQUE column behaves like a candidate key.
6. (C). NULL represents unknown/missing data.
7. (B) Relationship. Direct definitional recall.
8. (B) Book table. FK for a 1:N relationship sits on the “many” side.
9. (B) Determinant. In $X \rightarrow Y$, X is the determinant, Y is the dependent.
10. (D). A primary key value must always be non-NULL (entity integrity).

8.4 End-of-Day Quiz – Answers

1. (B). File systems have no built-in integrity enforcement or query language.
2. (B) Candidate key. Smallest possible super key that retains uniqueness.
3. (C) Referential integrity. A non-NULL FK pointing to a non-existent PK value.
4. (B) Cardinality. ER-specific meaning (distinct from row-count cardinality).
5. (B) RollNo $\rightarrow$ Name. Determinant on the left, dependent on the right.

9 Scoring and Diagnostic Interpretation

Score band	Interpretation and next action
<60%	Revisit today's concept section tonight before moving on. Focus especially on any topic where you missed 2+ questions.
60–75%	Acceptable. Mark the specific weak sub-topics in the error log below and do a 10-minute re-read.
75–85%	Good. Log any individual misses, then proceed as scheduled.
>85%	Strong. Proceed to the Exam Extension (Section 10) with confidence, then on to Day 2 (Normalization).

Note: This is a personal study diagnostic only, not an official IOCL cutoff or scoring benchmark. No IOCL-specific subject weightage is implied by these bands.

10 Day 1 Exam Extension — Functional Dependencies, Closure & Keys

Why this section exists

The core Day 1 session above deliberately keeps FD depth to a one-line preview (Section 2.8) – candidate-key/closure work was originally scheduled for later days. This optional 45–60 minute extension pulls forward only the specific FD/closure/key patterns that show up repeatedly in PSU and GATE papers, because they are high-yield, mechanical, and fast to learn once you know the steps. It is **not** a normalization chapter – see Section 10.11 for what is deliberately left out.

10.1 A. Functional Dependency Basics

A functional dependency $X \rightarrow Y$ means: whenever two tuples agree on X , they must agree on Y . X is the *determinant*, Y is the *dependent*.

- **Trivial FD:** $Y \subseteq X$ (e.g., $AB \rightarrow A$). Always true, carries no information.
- **Non-trivial FD:** $Y \not\subseteq X$ (e.g., $A \rightarrow B$).
- **Transitive dependency:** If $A \rightarrow B$ and $B \rightarrow C$, then C is transitively dependent on A through B – even if $A \rightarrow C$ could also be derived directly, the path through B is what makes it “transitive.”
- **Partial dependency:** Only meaningful when the candidate key is composite. If (A, B) is the candidate key and $A \rightarrow C$ holds, then C depends on only *part* of the key – a partial dependency.

Keep this at recognition/application level for now; the 2NF/3NF use of these ideas comes later (Section 10.11).

10.2 B. Attribute Closure (X^+)

Definition: X^+ (closure of X) is the set of all attributes functionally determined by X , given a set of FDs.

Mechanical Closure Algorithm

1. Start with **result** = X .
2. Repeat: for every FD $P \rightarrow Q$ in F , if $P \subseteq \mathbf{result}$, add Q to **result**.
3. Stop when a full pass adds no new attributes.
4. **result** is X^+ .

Worked example. $R(A, B, C, D, E)$, $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow A\}$. Find $(AD)^+$.

$$AD \rightarrow ABCD \text{ (via } A \rightarrow BC) \rightarrow ABCDE \text{ (via } CD \rightarrow E)$$

No new attribute can be added ($E \rightarrow A$ gives nothing new). So $(AD)^+ = \{A, B, C, D, E\}$ – the full relation, so AD is a **superkey**.

10.3 C. Candidate-Key Finding Using Closure

Algorithm

1. List every attribute of R .
2. Scan the RHS of every FD. Any attribute that **never** appears on any RHS **must** be in every candidate key (call this set the *core*).
3. Start from the core; if its closure already equals all of R , the core itself is a candidate key.
4. Otherwise, add one attribute at a time to the core, compute closure, and stop as soon as closure = all of R .
5. Check minimality: remove each attribute one at a time and confirm the closure shrinks (no longer covers all of R). If it still covers R , that attribute was unnecessary – drop it.

Worked example (same FDs as above). RHS attributes across $F = \{B, C, E, A\}$. Attribute D never appears on any RHS $\Rightarrow D$ is in every candidate key.

Try candidates containing D : AD , CD , ED .

$$AD^+ = ABCDE \quad (\checkmark \text{ superkey}), \quad A^+ = ABC \quad (\neq R), \quad D^+ = D \quad (\neq R)$$

Neither A alone nor D alone is a superkey, so AD is minimal $\Rightarrow$ **candidate key**.

Similarly $CD^+ = ABCDE$ and $ED^+ = ABCDE$, and neither C , E , nor D alone works alone $\Rightarrow$ CD and ED are also candidate keys.

Result: candidate keys are AD , CD , ED . (This is exactly the structure tested in GATE DA 2026, Q7 – see Section 10.12.)

10.4 D. Superkey vs Candidate Key

Mental Model

Every candidate key is a superkey. A candidate key is a **minimal** superkey – remove any attribute and it stops being a superkey.

$$\text{Superkeys} \supseteq \text{Candidate Keys}$$

If AB is a candidate key, then AB , ABC , ABD , $ABCD$, $\dots$ are all superkeys (every superset of a candidate key is automatically a superkey, because adding attributes cannot destroy uniqueness).

10.5 E. Counting Superkeys — ONE Candidate Key

Steps

1. Note n = total attributes in R ; k = size of the (single) candidate key.
2. The $n - k$ attributes *not* in the candidate key are each independently “include or don’t include” – 2 choices each.
3. Number of superkeys = 2^{n-k} .

Worked example. $R(A, B, C, D, E)$, candidate key = AB ($n = 5, k = 2$). Remaining attributes: C, D, E (3 attributes).

$$\text{Superkeys} = 2^{5-2} = 2^3 = 8$$

They are $AB, ABC, ABD, ABE, ABCD, ABCE, ABDE, ABCDE$.

10.6 F. Counting Superkeys — TWO Candidate Keys (Inclusion–Exclusion)

This is the case students most often get wrong, because superkeys containing *both* candidate keys get double-counted if you just add the two single-key totals.

Steps

1. Let K_1, K_2 be the two candidate keys, n = total attributes.
2. $|S_1| = 2^{n-|K_1|}$ = superkeys containing K_1 .
3. $|S_2| = 2^{n-|K_2|}$ = superkeys containing K_2 .
4. Find $K_1 \cup K_2$ (the union of the two key's attributes). Superkeys containing *both* keys are exactly the superkeys containing this union: $|S_1 \cap S_2| = 2^{n-|K_1 \cup K_2|}$.
5. Apply inclusion–exclusion: $|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|$.

Verified GATE PYQ — GATE CSE 2026, Set 1, Q55

$R(A, B, C, D)$. The only two candidate keys are $\{A, B\}$ and $\{A, C\}$. Find the number of superkeys. ($n = 4$)

$$|S_1| = 2^{4-|\{A,B\}|} = 2^{4-2} = 2^2 = 4 \quad (AB, ABC, ABD, ABCD)$$

$$|S_2| = 2^{4-|\{A,C\}|} = 2^{4-2} = 2^2 = 4 \quad (AC, ABC, ACD, ABCD)$$

$$K_1 \cup K_2 = \{A, B, C\}, \quad |S_1 \cap S_2| = 2^{4-3} = 2^1 = 2 \quad (ABC, ABCD)$$

$$\text{Total} = 4 + 4 - 2 = \boxed{6}$$

Answer verified: 6. This exact question and answer were confirmed directly from the official GATE CSE 2026 Set 1 paper (Q55, numerical-answer type) via GATE Overflow's question archive.

10.7 G. Counting Superkeys — THREE OR MORE Candidate Keys (Full Inclusion–Exclusion)

The same idea extends, but now every pairwise and triple-wise (and higher) overlap must be accounted for using the general inclusion–exclusion formula.

Steps for three candidate keys K_1, K_2, K_3

1. Compute each single-key term: $|S_i| = 2^{n-|K_i|}$ for $i = 1, 2, 3$.
2. Compute each pairwise-union term: $|S_i \cap S_j| = 2^{n-|K_i \cup K_j|}$ for every pair.
3. Compute the triple-union term: $|S_1 \cap S_2 \cap S_3| = 2^{n-|K_1 \cup K_2 \cup K_3|}$.
4. Combine with alternating signs:

$$|S_1 \cup S_2 \cup S_3| = \sum |S_i| - \sum |S_i \cap S_j| + |S_1 \cap S_2 \cap S_3|$$

Worked example. $R(A, B, C, D, E)$ ($n = 5$) has three candidate keys: AD , CD , ED (this is exactly the GATE DA 2026 Q7 result from Section 10.12 above).

$$\begin{aligned}
 |S_{AD}| &= 2^{5-2} = 8, & |S_{CD}| &= 2^{5-2} = 8, & |S_{ED}| &= 2^{5-2} = 8 \\
 AD \cup CD &= ACD \text{ (3 attrs)} \Rightarrow |S_{AD} \cap S_{CD}| = 2^{5-3} = 4 \\
 AD \cup ED &= ADE \text{ (3 attrs)} \Rightarrow |S_{AD} \cap S_{ED}| = 2^{5-3} = 4 \\
 CD \cup ED &= CDE \text{ (3 attrs)} \Rightarrow |S_{CD} \cap S_{ED}| = 2^{5-3} = 4 \\
 AD \cup CD \cup ED &= ACDE \text{ (4 attrs)} \Rightarrow |S_{AD} \cap S_{CD} \cap S_{ED}| = 2^{5-4} = 2 \\
 \text{Total} &= (8 + 8 + 8) - (4 + 4 + 4) + 2 = 24 - 12 + 2 = \boxed{14}
 \end{aligned}$$

Common Trap

Exam trap: Do not just add $|S_1| + |S_2| + |S_3|$ – every pairwise overlap is double-counted and the triple overlap is removed too many times and must be added back. This is the same logic as the set-theory inclusion–exclusion principle you already know from probability/counting: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

10.8 H. Prime and Non-Prime Attributes

Prime attribute: appears in *at least one* candidate key. **Non-prime attribute:** appears in *no* candidate key.

Example. $R(A, B, C, D, E)$, candidate keys AB and CD . Prime: A, B, C, D . Non-prime: E .

10.9 I. Composite-Key / Weak-Entity Reasoning

A composite key does not become a strong identifier merely because a foreign/owner attribute is added to a weak entity. A weak entity remains conceptually weak until its *owner's key* plus its own *partial key* together identify it.

Example. $\text{Dependent}(\text{EmployeeID}, \text{DependentName}, \dots)$. If one employee can have several dependents, **EmployeeID** alone is not unique within **Dependent** – it repeats once per dependent. The identifying combination is **EmployeeID** + **DependentName** (owner key + partial key). Adding **EmployeeID** alone does not make **Dependent** a strong entity.

10.10 J. Integrity-Constraint Clarification

- **PRIMARY KEY constraint** enforces *entity integrity*: uniqueness *and* non-nullability of the key attribute(s).

- **Referential integrity** concerns foreign-key references: a non-NULL FK value must match an existing parent-key value.
- A *nullable* FK may legitimately contain NULL (“no relationship yet”); a *non-NULL* FK must reference an existing parent row.

Don’t conflate the two: PK rules protect a row’s own identity; FK rules protect the *link* between two rows.

What Comes Later — Do Not Study Now

These are valid and important DBMS topics, but they are **not** covered in this extension and should not be studied today – they begin properly in later workbook days:

- Armstrong’s axioms and derived inference rules (union, decomposition, pseudotransitivity)
- Minimal / canonical cover
- Equivalence of FD sets
- Full normalization: 1NF, 2NF, 3NF
- BCNF
- Lossless-join decomposition
- Dependency preservation
- FD projection after decomposition
- Candidate keys after decomposition
- Multivalued dependencies / 4NF
- Join dependencies / 5NF

10.11 K. Exam Pattern Bank

Common Trap

Source labeling in this bank: Questions marked [Verified GATE PYQ] were confirmed against the official GATE question/answer archive. Questions marked [PSU/IOCL-pattern practice] are original practice questions written in the style commonly seen in PSU/IOCL-type papers – they are *not* claimed to be actual IOCL previous-year questions, since no IOCL-specific FD/closure PYQ could be independently verified from a public source at the time of writing.

P1 [Verified GATE PYQ – GATE CSE 2026, Set 1, Q55] [Superkey counting, multiple keys] [Hard]

$R(A, B, C, D)$; only candidate keys are $\{A, B\}$ and $\{A, C\}$. Number of superkeys?

Solution: See worked example in Section 10.6. Answer: **6**.

P2 [Verified GATE PYQ – GATE DA 2026, Q7] [Candidate-key finding, closure] [Medium]

$R(A, B, C, D, E)$, $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow A\}$. Which are the candidate keys?

Solution: See worked example in Section 10.3/10.7. Answer: AD, CD, ED .

P3 [PSU/IOCL-pattern practice] [Superkey counting, single key] [Easy]

$R(P, Q, R, S, T)$, candidate key = $\{P, Q\}$. How many superkeys exist?

Solution: $n = 5, k = 2 \Rightarrow 2^{5-2} = 8$.

P4 [PSU/IOCL-pattern practice] [FD validity] [Easy]

Table has columns (RollNo, City); two rows share RollNo=7 but have different City values. Does $\text{RollNo} \rightarrow \text{City}$ hold?

Solution: No – same RollNo must give the same City for the FD to hold; here it doesn’t.

P5 [PSU/IOCL-pattern practice] [Trivial vs non-trivial] [Easy]

Is $\{A, B\} \rightarrow B$ trivial or non-trivial?

Solution: $B \subseteq \{A, B\}$, so it is trivial.

P6 [PSU/IOCL-pattern practice] [Closure] [Medium]

$R(A, B, C, D)$, $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. Find A^+ .

Solution: $A \rightarrow AB \rightarrow ABC \rightarrow ABCD$. $A^+ = \{A, B, C, D\}$, so A alone is a candidate key.

P7 [PSU/IOCL-pattern practice] [Multiple candidate keys, RHS shortcut] [Hard]

$R(A, B, C, D)$, $F = \{A \leftrightarrow B, C \leftrightarrow D\}$ (meaning $A \rightarrow B, B \rightarrow A, C \rightarrow D, D \rightarrow C$). Find all candidate keys.

Solution: RHS attributes = $\{A, B, C, D\}$ (every attribute appears on some RHS), so no attribute is forced into every key by the shortcut – try combinations directly. $(AC)^+ = ABCD$; minimal since $A^+ = AB \neq R$, $C^+ = CD \neq R$. By symmetry AD, BC, BD are also candidate keys. All four: AC, AD, BC, BD .

P8 [PSU/IOCL-pattern practice] [Superkey counting, three keys] [Hard]

$R(A, B, C, D, E)$ has exactly the candidate keys AD, CD, ED (from P2 above). How many superkeys in total?

Solution: See worked example in Section 10.7. Answer: **14**.

P9 [PSU/IOCL-pattern practice] [Prime/non-prime] [Easy]

$R(A, B, C, D, E)$, candidate keys AB and CD . List prime and non-prime attributes.

Solution: Prime: A, B, C, D . Non-prime: E .

P10 [PSU/IOCL-pattern practice] [Weak entity] [Medium]

$\text{Dependent}(\text{EmployeeID}, \text{DependentName}, \text{DOB})$. One employee has several dependents. Is EmployeeID alone a valid key for Dependent ?

Solution: No – it repeats for each dependent of the same employee. The identifying key is $(\text{EmployeeID}, \text{DependentName})$.

P11 [PSU/IOCL-pattern practice] [Integrity constraints] [Easy]

A nullable foreign key column currently holds NULL in one row. Is referential integrity violated?

Solution: No. A nullable FK is allowed to be NULL; referential integrity is only violated by a non-NULL FK value with no matching parent key.

P12 [PSU/IOCL-pattern practice] [Superkey vs candidate key] [Easy]

If AB is the only candidate key of $R(A, B, C)$, is ABC a candidate key?

Solution: No – ABC is a superkey (it's a superset of AB) but not minimal, since AB alone already determines everything.

P13 [PSU/IOCL-pattern practice] [Transitive dependency] [Medium]

$F = \{\text{EmpID} \rightarrow \text{DeptID}, \text{DeptID} \rightarrow \text{DeptName}\}$. Is DeptName transitively dependent on EmpID ?

Solution: Yes – DeptName depends on EmpID only via the intermediate attribute DeptID , which is the definition of transitive dependency, regardless of whether $\text{EmpID} \rightarrow \text{DeptName}$ can also be inferred directly.

11 Day 1 Extension Practice (15 Questions)

Difficulty mix: $\approx 30\%$ easy, 50% medium, 20% hard. At least 5 questions need closure/key reasoning; at least 2 involve multiple candidate keys or superkey counting; at least one weak-entity question and one integrity-constraint question are included.

Worked Examples (study these first)

WE1 [Easy] [Closure] $R(A, B, C, D)$, $F = \{A \rightarrow B, B \rightarrow C\}$. Find A^+ .

$$A \rightarrow AB \quad (A \rightarrow B) \rightarrow ABC \quad (B \rightarrow C)$$

No FD adds D . $A^+ = \{A, B, C\}$ (not the full relation, since D is missing) $\Rightarrow A$ is *not* a superkey by itself.

WE2 [Medium] [Candidate key via closure] $R(A, B, C, D)$, $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. Find the candidate key. RHS attributes = $\{B, C, D\}$; A never appears on RHS $\Rightarrow A$ must be in every candidate key.

$$A \rightarrow AB \rightarrow ABC \rightarrow ABCD$$

$A^+ = \{A, B, C, D\} = R$, and A is a single attribute so it's automatically minimal. **Candidate key:** A .

WE3 [Hard] [Superkey counting, two keys] $R(W, X, Y, Z)$, candidate keys $\{W, X\}$ and $\{W, Y\}$. Count the superkeys.

$$|S_1| = 2^{4-2} = 4 \quad (WX, WXY, WXZ, WXYZ)$$

$$|S_2| = 2^{4-2} = 4 \quad (WY, WXY, WYZ, WXYZ)$$

$$\text{Overlap } K_1 \cup K_2 = \{W, X, Y\}, \quad |S_1 \cap S_2| = 2^{4-3} = 2 \quad (WXY, WXYZ)$$

$$\text{Total} = 4 + 4 - 2 = \boxed{6}$$

(The overlap $\{WXY, WXYZ\}$ is counted once in $|S_1|$ and once in $|S_2|$ – shown explicitly above – so it must be subtracted once.)

Guided Questions (with hints)

G1 [Easy] [Closure] $R(A, B, C, D, E)$, $F = \{A \rightarrow BC, C \rightarrow D\}$. Find A^+ .

Hint: chain through both FDs; note E is never reachable.

G2 [Medium] [Candidate key] $R(A, B, C, D)$, $F = \{AB \rightarrow C, C \rightarrow D\}$. Is AB a candidate key?

Hint: compute $(AB)^+$ and check both minimality conditions (A^+ , B^+ alone).

G3 [Medium] [Superkey counting, one key] $R(P, Q, R, S)$, candidate key = $\{P\}$. How many superkeys?

Hint: $n = 4, k = 1$.

G4 [Hard] [Superkey counting, two keys] $R(A, B, C, D, E)$, candidate keys $\{A, B\}$ and $\{C, D\}$ (disjoint keys, no shared attribute). How many superkeys?

Hint: since the keys are disjoint, $K_1 \cup K_2$ has 4 attributes – compute the overlap term carefully before subtracting.

G5 [Easy] [Integrity constraint] A FK column is declared NOT NULL and currently every row has a valid matching parent-key value. A new row is inserted with FK = NULL. What happens?

Hint: distinguish the NOT NULL constraint from referential integrity itself.

Independent Questions

I1 [Easy] [Trivial/non-trivial] $R(A, B, C)$, is $AB \rightarrow B$ trivial or non-trivial?

I2 [Easy] [Prime/non-prime] $R(A, B, C, D)$, candidate keys $\{A\}$ and $\{B\}$. List prime and non-prime attributes.

I3 [Medium] [Closure] $R(A, B, C, D, E)$, $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E\}$. Find $(AB)^+$.

I4 [Medium] [Candidate key, RHS shortcut] $R(A, B, C, D, E)$, $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E\}$. Find the candidate key.

I5 [Medium] [Weak entity] A Room entity is weak, identified within a Building. Attributes: BuildingID, RoomNumber, Capacity. What is the identifying key of Room, and why isn't BuildingID alone sufficient?

I6 [Hard] [Superkey counting, two keys] $R(A, B, C, D, E, F)$ ($n = 6$), candidate keys $\{A, B\}$ and $\{A, C\}$. How many superkeys?

I7 [Hard] [Candidate key, multiple keys] $R(A, B, C, D)$, $F = \{AB \rightarrow CD, CD \rightarrow AB\}$. Find all candidate keys.

Answer Key with Full Solutions

G1. $A \rightarrow ABC$ ($A \rightarrow BC$) $\rightarrow ABCD$ ($C \rightarrow D$). $A^+ = \{A, B, C, D\}$; E is unreachable, so A is *not* a superkey.

G2. $(AB)^+$: start $AB \rightarrow ABC$ ($AB \rightarrow C$) $\rightarrow ABCD$ ($C \rightarrow D$) = R . So AB is a superkey. Minimality: $A^+ = A$ (no FD fires, since AB is needed together) $\neq R$; $B^+ = B \neq R$. Neither alone works $\Rightarrow$ **yes, AB is a candidate key.**

G3. $n = 4, k = 1 \Rightarrow 2^{4-1} = 2^3 = 8$ superkeys.

G4. $|S_1| = 2^{5-2} = 8$ (containing AB), $|S_2| = 2^{5-2} = 8$ (containing CD). $K_1 \cup K_2 = \{A, B, C, D\}$ (4 attributes, since the keys are disjoint) $\Rightarrow |S_1 \cap S_2| = 2^{5-4} = 2$. Total = $8 + 8 - 2 = 14$.

G5. The insert is **rejected** by the NOT NULL constraint itself – this is a domain/column-level constraint, not referential integrity. (Referential integrity governs what happens when the FK holds a *non-NULL* value that doesn't match a parent row; here the NOT NULL rule blocks the NULL before referential integrity even comes into play.)

I1. $B \subseteq AB$, so $AB \rightarrow B$ is **trivial**.

I2. Prime: A, B . Non-prime: C, D .

I3. $AB \rightarrow ABC$ ($AB \rightarrow C$) $\rightarrow ABCD$ ($C \rightarrow D$) $\rightarrow ABCDE$ ($D \rightarrow E$). $(AB)^+ = \{A, B, C, D, E\} = R$.

I4. From I3, $(AB)^+ = R$, so AB is a superkey. RHS attributes across $F = \{C, D, E\}$; A and B never appear on any RHS, so *both* must be in every candidate key – confirming AB (not a smaller set) is required. Minimality: $A^+ = A \neq R$, $B^+ = B \neq R$. **Candidate key: AB .**

I5. Identifying key: (**BuildingID**, **RoomNumber**). **BuildingID** alone is not sufficient because many rooms share the same building – it repeats across all rows for that building. **RoomNumber** is the partial key that, combined with the owner's key **BuildingID**, uniquely identifies each room.

I6. $|S_1| = 2^{6-2} = 16$ (AB), $|S_2| = 2^{6-2} = 16$ (AC). $K_1 \cup K_2 = \{A, B, C\}$ (3 attrs) $\Rightarrow |S_1 \cap S_2| = 2^{6-3} = 8$. Total = $16 + 16 - 8 = 24$.

I7. $(AB)^+$: $AB \rightarrow ABCD$ (via $AB \rightarrow CD$) = R , so AB is a superkey; $A^+ = A \neq R$, $B^+ = B \neq R$, so AB is minimal $\Rightarrow$ candidate key. By symmetry, $(CD)^+ = CD \rightarrow ABCD$ (via $CD \rightarrow AB$) = R , and $C^+ = C \neq R$, $D^+ = D \neq R$, so CD is also a candidate key. **Candidate keys: AB and CD .**

12 Revision Checklist

Core Day 1 Checklist

- ☐ I can state the difference between DBMS and RDBMS in one sentence.
- ☐ I can define schema vs instance without hesitation.
- ☐ I can name all 4 basic relational terms: relation, tuple, attribute, domain.
- ☐ I can order the key hierarchy: super key $\supseteq$ candidate key $\supseteq$ primary key.
- ☐ I can identify a composite key from a sample table.
- ☐ I can distinguish entity integrity from referential integrity.
- ☐ I know a nullable foreign key CAN legally be NULL.
- ☐ I know $\text{NULL} \neq 0 \neq \text{''}$ and that NULL propagates through arithmetic/comparisons.
- ☐ I can identify 1:1, 1:N, and M:N cardinalities from a short scenario.
- ☐ I understand $X \rightarrow Y$ means “X determines Y” (without diving into closures yet).

Exam Extension Checklist (added 19 August 2026)

- ☐ I can calculate X^+ for a simple FD set.
- ☐ I can use RHS analysis to identify attributes that must occur in every candidate key.
- ☐ I can find and verify candidate keys using closure and minimality.
- ☐ I can distinguish a candidate key from a superkey.
- ☐ I can count superkeys when there is one candidate key.
- ☐ I can count superkeys with two candidate keys without double-counting.
- ☐ I can count superkeys with three or more candidate keys using full inclusion–exclusion.
- ☐ I can identify prime and non-prime attributes.
- ☐ I can explain why EmployeeID + DependentName identifies a weak entity while EmployeeID alone does not.

Things I Got Wrong Today

Question	Topic	My mistake	Correct rule	Revisit date
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13 One-Page Cheat Sheet

DBMS vs RDBMS

DBMS = software to manage data. RDBMS = DBMS + strict tables + enforced keys/constraints. Every RDBMS is a DBMS; not every DBMS is an RDBMS.

Schema vs Instance

Schema = fixed design/structure (rare changes). Instance = actual data right now (frequent changes).

Core Terms

Relation = table. Tuple = row. Attribute = column. Domain = valid value set. Degree = #columns. Cardinality = #rows.

Key Hierarchy

Super key $\supseteq$ Candidate key $\supseteq$ Primary key. Alternate key = candidate key not chosen as PK. Composite key = key spanning 2+ attributes. Foreign key = links to another table's PK.

Integrity Constraints

Domain constraint: valid value/type. Entity integrity: PK never NULL. Referential integrity: FK matches PK or is NULL. Key constraint: candidate key values unique.

ER Cardinalities

1:1 – at most one on each side. 1:N – FK goes on the “many” side. M:N – needs a separate junction table.

NULL Basics

NULL = unknown/missing, not 0, not “. Arithmetic/-comparison with NULL $\rightarrow$ NULL. COUNT(*) counts all rows; COUNT(col) skips NULLs.

FD Intuition

$X \rightarrow Y$: “X determines Y.” Same X value $\Rightarrow$ same Y value across tuples.

Exam Extension Formulas

Closure algorithm: start with X; repeatedly add RHS of any FD whose LHS $\subseteq$ current closure; stop when nothing new is added.

Candidate key shortcut: attributes never on any RHS $\Rightarrow$ must be in every candidate key. Build up from there, test closure, confirm minimality.

Superkey count, one CK of size k (n attrs total):

$$2^{n-k}$$

Superkey count, two CKs K_1, K_2 :

$$2^{n-|K_1|} + 2^{n-|K_2|} - 2^{n-|K_1 \cup K_2|}$$

Superkey count, three CKs K_1, K_2, K_3 : full inclusion-exclusion –

$$\sum 2^{n-|K_i|} - \sum 2^{n-|K_i \cup K_j|} + 2^{n-|K_1 \cup K_2 \cup K_3|}$$

Prime attribute: in ≥ 1 candidate key. **Non-prime:** in no candidate key.

Weak entity: owner's key + partial key = identifying key (owner key alone is not enough if the child repeats per owner).

End of Day 1 Workbook (Core + Exam Extension) – IOCL Grade-A 2026 CS/IT Preparation

Next: Day 2 – DBMS Normalization (Armstrong's axioms, minimal cover, 1NF–3NF, BCNF)